Explanations That Arrive in Time: Deadline-Aware, Control-Loop-Structured Explainable AI for Intrusion Detection in Cyber-Physical Systems

Md Mazharul Islam[a,*], Sazzad Hossain[b,1]

[a]*North South University, Dhaka, Bangladesh*
[b]*Department of Electrical and Computer Engineering, North South University, Bashundhara, Dhaka, 1229, Bangladesh*

Abstract

Explainable AI (XAI) is proposed to help operators act on intrusion alarms in cyber-physical systems (CPS), yet evaluations rarely test whether explanations identify the compromised part of the process in time. No prior work relates explanation correctness to how long the process tolerates an attack. We investigate whether, and with which design, explanations of machine-learning intrusion detectors localize the attacked control loop before a physically derived deadline. We define deadline-conditioned metrics combining detection delay, explainer latency and correctness, curate loop-level ground truth for all 50 HAI 21.03 attacks, build a closed-loop simulator with 280 hazard-timed attacks, and compare 13 configurations, including exact loop Shapley values (LX), physics-informed attribution (PI-CAX) and temporal evidence accumulation (TEA), across four detector families with leakage-free protocols. Score attributions degrade as attacks propagate through coupled loops (HAI: LX falls from 0.79 to 0.60 top-1 accuracy within 60 s); TEA recovers only part of this loss. PI-CAX, which separates the manipulated loop's persistent departure from compensating oscillations, significantly outperforms every other configuration: on HAI it reaches 0.93 top-1 accuracy at 60 s and localizes the attacked loop within 60 s of onset for 0.72 of attacks, versus at most 0.54 for others. In simulation, 86 % of deadline misses stem from late or missed detection. Grounding CPS explanations in control-loop structure and physical priors improves their timely accuracy, while timely detection remains the binding constraint.

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*Corresponding author.

Email addresses: mazharul.islam1@northsouth.edu (Md Mazharul Islam), hossain.sazzad01@northsouth.edu (Sazzad Hossain), sazzad.hossain.cs@gmail.com (Sazzad Hossain)

[1]ORCID: 0000-0002-9675-0468.

control loops, physics-informed explanation, process safety time

1. Introduction

Cyber-physical systems (CPS) couple computation with physical processes through sensing, control and actuation, so attacks on them have physical consequences. Machine-learning anomaly detectors are a widely studied defence: a forecaster learns the normal dynamics of sensors, setpoints and actuator commands and raises an alarm when observations depart from its predictions (Kravchik and Shabtai, 2018; Deng and Hooi, 2021; Fung et al., 2024). An alarm alone does not tell the operator *what* to do. The operator must know which part of the process has been manipulated, and the natural unit of a CPS is the feedback control loop: a setpoint (SP), a measured process variable (PV) and a control output (CV) tied together by one controller, which is also the unit in which industrial attacks are specified (Shin et al., 2021). Post-hoc attribution methods such as SHAP (Lundberg and Lee, 2017), LIME (Ribeiro et al., 2016) or integrated gradients (Sundararajan et al., 2017) are the usual answer.

Two observations motivate this work. First, the most systematic study of attribution for industrial anomaly detection found that attributions computed when the alarm fires are often wrong (Fung et al., 2024). Second, studies that measure the cost of explanations in CPS report it in absolute terms: seconds per explanation on a GPU (Oinonen and Morsi, 2026), time per window against a one-second budget (Franco de la Peña et al., 2026), or milliseconds on an edge device (Ericson et al., 2025). None relates the explanation, and its correctness at the time it is delivered, to what bounds the operator's response: the time the process tolerates the attack before reaching an unsafe state, formalised in process-safety engineering as the *process safety time* (International Electrotechnical Commission, 2016). The question that distinguishes XAI for CPS from XAI for IT intrusion detection is therefore: *does the operator receive a correct localization before the deadline imposed by the process?*

Approach. We evaluate explanations at the level of control loops and against deadlines. We compare two loop-structured explainers built on opposite design principles. LX computes exact Shapley values of the detector score over control loops, with an incident-local reference and a fixed cost of 2^L detector queries; it is axiomatic and agnostic to process physics. PI-CAX, developed in an

earlier phase of this project on synthetic plant models and evaluated here unchanged, encodes a physical hypothesis instead: a manipulated loop departs persistently from its normal input–output relations, whereas loops that merely react to it oscillate. Both are combined with temporal evidence accumulation (TEA), a method-agnostic aggregation of attributions over an incident.

Contributions.

1. **Deadline-conditioned evaluation.** The metrics DCL@$k(D)$ and the in-time rate against physical hazard deadlines account for detection delay, explainer latency and the correctness of the explanation actually delivered (Section 3).
2. **Benchmarks with localization ground truth.** We release loop- and tag-level ground truth for all 50 attacks of HAI 21.03 (Shin et al., 2021), cross-checked against an independent transcription (49/50 agreement), and PCL-Sim, an open closed-loop simulator whose 280 attacks carry hazard deadlines computed from the true physical state (Section 5).
3. **Loop-structured explainers and TEA** (Section 4), with an ablation identifying which physical prior makes PI-CAX work.
4. **A leakage-free, statistically tested comparison** of 13 attribution configurations, four detector families and three seeds on both benchmarks, covering accuracy over time, deadlines, latency, faithfulness, stability, noise and decoy robustness (Section 6).

Summary of findings.

- *Score attributions fail as attacks propagate.* Every gradient-, sampling- or Shapley-based method degrades after the alarm as controllers spread the effect to coupled loops. TEA recovers part of the loss for all of them, after which the leading score attributions are within a few points of each other.

- *A physical prior fixes this.* PI-CAX is significantly more accurate than every other configuration on both benchmarks, at the alarm and after it. Almost all of its advantage comes from separating persistent from oscillatory deviations.

- *Loop structure is essential.* Data-driven or random groupings fail. In HAI, manipulated setpoints almost never appear in the recorded data (2 of 36 cases), so only the loop, not the tag, can be identified.

- *Detection limits timeliness.* On hazardous simulated attacks, 86 % of deadline misses are late or missed detections.

- *Leakage matters.* A random window split inflates supervised detection F1 on HAI from 0.575 to 0.940.

Paper organization. The remainder of this paper is organized as follows. Section 2 reviews related work on XAI for CPS intrusion detection and on evaluating explanations. Section 3 formalizes the system and threat model and defines the deadline-conditioned metrics. Section 4 presents LX, TEA, tag refinement and PI-CAX. Section 5 describes the benchmarks, detectors, explainers and evaluation protocol. Section 6 reports the results, Section 7 discusses their implications and limitations, and Section 8 concludes.

2. Related work

Interpretable and post-hoc XAI for CPS intrusion detection. Some detectors expose interpretable internals: graph-attention weights that point to a deviating sensor (Deng and Hooi, 2021; Tang et al., 2023), attention on ICS traffic (Eltomy and Lalouani, 2024), rules distilled from a logical neural network (Xu et al., 2024), or domain-aware relevance propagation (Rasheed et al., 2026). Most of the literature applies SHAP or LIME to a detector or classifier, for SCADA and gas-pipeline autoencoders (Huong et al., 2022; Ha et al., 2022), federated and correlation-based IIoT detectors (Yazdinejad et al., 2025; Birihanu and Lendák, 2024; Abudurexiti et al., 2025), microgrids (Trivedi et al., 2025), process-oriented attacks (Kenmogne and Mocanu, 2024, 2025) and network-traffic IIoT detectors (Oseni et al., 2023; Khan et al., 2022; Shoukat et al., 2025; Alani, 2023; Attique et al., 2024; Ahakonye et al., 2024; Saiyed et al., 2026), or uses surrogate and example-based explanations (Zolanvari et al., 2023; Panesar et al., 2026). Interpretability is mostly validated qualitatively or by expert agreement. Several studies evaluate time-series data with random proportional splits or k-fold cross-validation (Supplementary Table S1), a known source of optimistic bias (Arp et al., 2022; Kaufman et al., 2012).

Evaluating explanations of intrusion detectors. Warnecke et al. (2020) proposed security-specific criteria (accuracy, sparsity, stability, efficiency, robustness, completeness), and Wei et al. (2023) and Arreche et al. (2024) evaluate network-IDS explanations along similar axes. Oinonen and Morsi (2026)

compare seven XAI methods for a substation IDS, and WaXAI (Mathuros et al., 2024) measures overlap between top-ranked features and attacked SWaT devices. Fung et al. (2024) provide the most systematic ground-truth evaluation: across 156 manipulations on SWaT, WADI and a Tennessee Eastman simulation, attributions at practical detection timings are often inaccurate, and an ensemble of raw-error, saliency and LEMNA (Guo et al., 2018) attributions helps. We build on their protocol, change the unit of attribution to the control loop, follow explanations over the incident, and add timeliness.

Grouped, temporal and counterfactual explanations. Owen values (Owen, 1977) extend Shapley values (Shapley, 1953) to a priori unions of players, and sampling estimators make them tractable (Castro et al., 2009; Štrumbelj and Kononenko, 2014; Lundberg and Lee, 2017). ShaTS (Franco de la Peña et al., 2026) groups time-series features by time step, sensor or SWaT process stage and reports execution time against a one-second budget; our loop game is an instance of such a priori grouping, with control loops as groups and an incident-local reference. Mishra et al. (2026) compute sensor–time attributions with conditional counterfactuals for root-cause analysis. AR-Pro (Ji et al., 2024) produces diffusion-based counterfactual repairs with formal properties, evaluated on SWaT, WADI and HAI, and JoCE (Jeong et al., 2026) searches for sparse joint time–feature edits. Counterfactuals answer what would have to change rather than which loop is responsible; none of these methods has been evaluated against documented targets under deadlines.

Timeliness in CPS security. Cárdenas et al. (2011) and Urbina et al. (2016) analyse attacks and detection in terms of physical impact, and Puccetti and Ceccarelli (2026) argue that detection latency must be evaluated together with false-positive rates. GiBy (Venugopalan and Adepu, 2025) explains detections through per-sensor safety bounds within milliseconds. To our knowledge, no prior work evaluates whether a *correct* explanation reaches the operator within a process-derived deadline. Table 1 summarises the closest work.

3. Problem formulation and threat model

System model. A historian or SCADA feed delivers process-level tags $\boldsymbol{x}_t \in \mathbb{R}^d$ at a fixed sampling period (1 s in both benchmarks). The tags are partitioned into L control loops $\mathcal{G} = \{g_1, \ldots, g_L\}$, where $g_\ell \subseteq \{1, \ldots, d\}$ contains the setpoint, process variable(s), controller output and actuator feedback of loop

Table 1: Positioning against the closest prior work ("–": not reported).

	Fung et al. (2024)	ShaTS (Franco de la Peña et al., 2026)	Oinonen and Morsi (2026)	XAI-CPS (Rasheed et al., 2026)	**This work**
Attribution unit	tag	tag / process stage	pixel	tag	control loop (+tag refinement)
Ground-truth localization	yes	case-based	no	expert agreement	yes (2 benchmarks, cross-checked)
At operational timing	yes (fails)	–	n/a	–	primary setting
Across alarm windows	no	no	no	no	yes (TEA)
Physical prior in explainer	no	no	no	physics-informed LRP	yes (PI-CAX)
Physical deadline	no	no	no	no	yes
Statistical tests	ANOVA/ Pearson	–	–	mean±sd	Friedman, Wilcoxon–Holm, bootstrap

ℓ. Tags not owned by a controller (e.g. measured disturbances or simulator states) form auxiliary groups. The partition is engineering metadata, available from the piping-and-instrumentation diagram or controller configuration; it is fixed before any test data is seen.

Detector. The detector is a forecaster f that maps a window $W_t = \boldsymbol{x}_{t-w:t-1}$ to $\hat{\boldsymbol{x}}_t = \boldsymbol{x}_{t-1} + f(W_t)$. For a segment $Z = [W_t; \boldsymbol{x}_t] \in \mathbb{R}^{(w+1)\times d}$ the anomaly score is

$$e(Z) = \frac{1}{d}\sum_{j=1}^{d} \psi\left(\frac{x_{t,j} - \hat{x}_{t,j}}{\sigma_j}\right)^2, \qquad \psi(r) = c\tanh(r/c), \tag{1}$$

where σ_j is the residual standard deviation on held-out normal data (floored at 0.05) and ψ is a differentiable soft clip ($c = 20$). An alarm is raised when the exponentially weighted score $s_t = \alpha e_t + (1-\alpha)s_{t-1}$ exceeds θ, the q-quantile of s on chronologically held-out normal validation data.

Threat model.

- *Access.* The adversary has gained access to the control network (Purdue levels 1–2). Over an interval $[t_{\text{on}}, t_{\text{off}})$ the adversary can: (a) change a setpoint while hiding the change from the HMI; (b) override or offset an actuator command; (c) spoof a sensor value by bias, freezing or replay, as seen by both the controller and the historian; or (d) combine these, e.g. a CV attack masked by PV replay, on one or two loops. This matches the I/O-point attack taxonomy of HAI (Shin et al., 2021).

- *Goal.* Drive the process toward an unsafe state or disrupt production.
- *Knowledge.* In the main experiments the adversary does not know the detector or the explainer. In the decoy experiment (RQ4) the adversary knows that attributions are shown to operators. In addition to the main attack, the adversary then applies a small bias to the PV of an innocent loop to shift blame.
- *Out of scope.* Network-layer attacks without process effect, compromise of the IDS host, and training-data poisoning.
- *Defender's objective.* Deliver the identity of the compromised loop(s), ideally with the manipulated tags, before the process safety time is consumed.

Deadline-conditioned localization. Consider an attack i with onset $t_{\mathrm{on},i}$, target loops $\mathcal{G}_i^\star$ and first alarm $t_{a,i}$ (undefined if missed). An explainer with latency ℓ produces explanations sequentially. It starts at the alarm and computes one explanation every $\Delta_{\mathrm{eff}} = \Delta\lceil \max(\Delta, \ell)/\Delta \rceil$ seconds, where Δ is the nominal refresh period. The j-th explanation uses data up to $t_{a,i} + j\Delta_{\mathrm{eff}}$ and becomes available at $t_{a,i} + j\Delta_{\mathrm{eff}} + \ell$. Let $\boldsymbol{\rho}_i(D) \in \mathbb{R}^L$ be the loop ranking available at $t_{\mathrm{on},i} + D$, i.e. the last explanation delivered by then.

Definition 1 (DCL)**.** *The deadline-conditioned localization at depth k is*

$$\mathrm{DCL@}k(D) = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}\Big[t_{a,i} \text{ exists}, \ \exists \text{ delivery by } t_{\mathrm{on},i} + D, \ \mathcal{G}_i^\star \cap \mathrm{top}_k\, \boldsymbol{\rho}_i(D) \neq \emptyset\Big]. \quad (2)$$

Missed detections, explanations that are not ready, and wrong explanations all count as failures. Hence DCL@k is upper-bounded by the event recall of the detector.

Definition 2 (In-time rate)**.** *When the hazard deadline D_i^{haz} of each attack is known, the in-time rate is $\frac{1}{N_h}\sum_{i:\, D_i^{\mathrm{haz}} < \infty} \mathbb{1}[\cdot]$ with $D = D_i^{\mathrm{haz}} - t_{\mathrm{act}}$ per attack. Here D_i^{haz} is the time from onset until the* true *physical state first violates a safety limit with no operator response, and t_{act} is a reserve for the operator's action.*

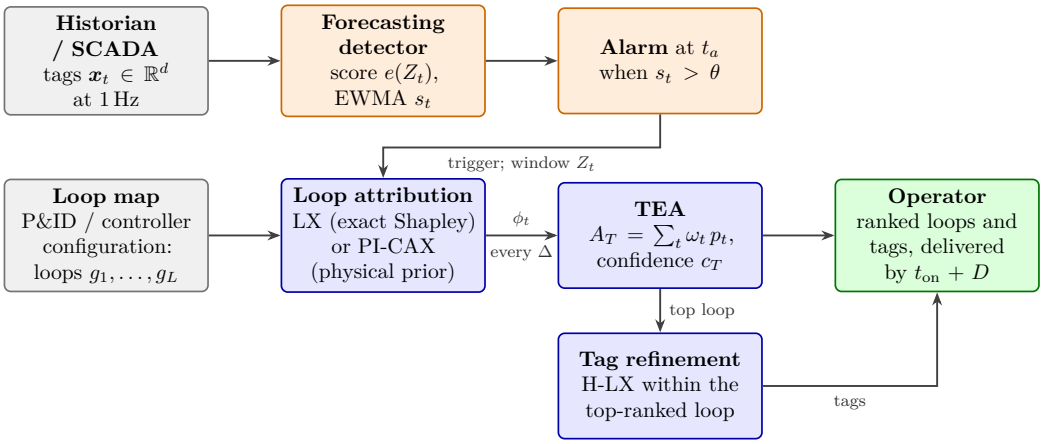


Figure 1: Deadline-aware explanation pipeline. Grey: inputs; orange: detection; blue: explanation components studied in this paper (LX and PI-CAX are alternative loop-level explainers); green: delivery to the operator. After the alarm, the selected explainer attributes the anomaly to control loops every Δ seconds, TEA accumulates these attributions, and the top loop is refined to tags. Detection delay, explainer latency and correctness together determine whether a correct localization reaches the operator before the deadline D.

D_i^{haz} is the empirical counterpart of the process safety time of IEC 61511 (International Electrotechnical Commission, 2016). It can be computed in simulation but is not available for recorded datasets. On HAI we therefore report DCL over a grid of deadlines.

4. Method

Figure 1 summarises the pipeline. The detector raises an alarm; from then on, a loop-level explainer (LX or PI-CAX) explains the anomaly every Δ seconds; TEA accumulates these explanations into a ranking with a confidence margin; and the top loop is refined to tags.

4.1. Loop game with an incident-local reference

At alarm time t_a we fix a last-known-good reference time $t_{\text{ref}} = t_a - \Delta_{\text{ref}}$ ($\Delta_{\text{ref}} = 60\,\text{s}$ by default; ablated in Section 6.7). Let $Z_t = \boldsymbol{x}_{t-w:t}$ be the current segment and $R = \boldsymbol{x}_{t_{\text{ref}}-w:t_{\text{ref}}}$ the reference segment. For a coalition of loops

$S \subseteq \mathcal{G}$ define the hybrid segment

$$[Z_t^S]_{\tau j} = \begin{cases} [Z_t]_{\tau j} & j \in \bigcup_{g \in S} g, \\ [R]_{\tau j} & \text{otherwise}, \end{cases} \tag{3}$$

and the characteristic function $v_t(S) = e(Z_t^S) - e(Z_t^\emptyset)$.

The game asks how much of the current anomaly is explained by each loop departing from its last known good trajectory. The common interventional alternative averages over training background windows (Lundberg and Lee, 2017; Franco de la Peña et al., 2026), which come from other operating points and can make hybrids look anomalous themselves; we compare both choices (LX vs. LX-bg).

*4.2. Exact loop Shapley values (*LX*)*

The attribution of loop g at time t is its Shapley value (Shapley, 1953)

$$\phi_t(g) = \sum_{S \subseteq \mathcal{G} \setminus \{g\}} \frac{|S|!\,(L - |S| - 1)!}{L!} \Big[v_t(S \cup \{g\}) - v_t(S) \Big]. \tag{4}$$

Proposition 1. LX *requires exactly* 2^L *evaluations of* $e(\cdot)$ *per explanation, independent of the data and of* d*. It satisfies efficiency,* $\sum_g \phi_t(g) = e(Z_t) - e(R)$*. Since it is exact, it also satisfies the symmetry, dummy and additivity axioms.*

Proof. All 2^L coalitions are evaluated once, and Eq. (4) is a fixed linear map of these values. The axioms are those of the Shapley value (Shapley, 1953), which hold because no sampling approximation is involved. □

The 2^L hybrid segments are evaluated in one batch (≤ 1024 forward passes for $L \leq 10$), so latency is *deterministic and known before deployment*, unlike sampling estimators (Castro et al., 2009; Štrumbelj and Kononenko, 2014); exact Shapley values over d tags would need 2^d evaluations ($d = 57$ on HAI). Loop grouping is an a priori union in the sense of Owen (1977), and it keeps a loop's SP, PV, CV and feedback together in every coalition, addressing the sensor–actuator coupling that Fung et al. (2024) identify as a cause of attribution failure.

*4.3. Temporal evidence accumulation (*TEA*)*

An incident produces a sequence of explanations at times $t_a, t_a + \Delta, \ldots$. Single-shot attributions are noisy. Worse, as the attack propagates through coupled loops, later residuals increasingly appear in loops that merely *react*. Following Fung et al. (2024), we expect the earliest windows to be most informative but not always reliable. TEA therefore aggregates normalised positive attributions, weighted by severity:

$$p_t(g) = \frac{\max(\phi_t(g), 0)}{\sum_h \max(\phi_t(h), 0)}, \qquad A_T(g) = \sum_{t=t_a}^{T} \omega_t \, p_t(g), \qquad \omega_t = \log^+ \frac{s_t}{\theta} + \varepsilon. \tag{5}$$

The ranking published at T sorts loops by A_T, with confidence margin $c_T = (A_T^{(1)} - A_T^{(2)}) / \sum_g A_T(g)$, where $A_T^{(r)}$ is the r-th largest accumulated value. A deployment may publish as soon as $c_T \geq \gamma$, or at the deadline, whichever is first. TEA is agnostic to the attribution method, so we apply it to every baseline as well.

*4.4. Tag refinement (H-*LX*)*

Operators eventually need tags. Within the top-ranked loop $g^\star$ we compute exact Shapley values over its $|g^\star|$ tags. All other loops are held at their current values, so the game isolates within-loop responsibility; the cost is $2^{|g^\star|}$ evaluations ($\leq 2^{12}$ on HAI).

4.5. Physics-informed loop attribution (PI-CAX)

LX is axiomatic: it distributes the detector score without any knowledge of how control loops behave. As a second, contrasting design we evaluate PI-CAX, a physics-informed causal attribution developed in an earlier phase of this project on synthetic multi-stage plant models and used here *unchanged* (all constants as designed; none was tuned on HAI or PCL-Sim). It encodes one physical hypothesis: a manipulated loop shows a *persistent, one-sided* departure from its normal input–output relations, whereas loops that merely react to it show *oscillatory* compensation. At time t it uses the last ten samples and proceeds in five steps.

1. *Detector evidence.* $z^{\mathrm{det}} = 0.65|z_t| + 0.25|z_{t-1}| + 0.10|z_{t-2}|$, where z_τ are the standardised forecaster residuals.

Algorithm 1 Deadline-aware localization for one incident

Require: detector (f, σ, θ), loops $\mathcal{G}$, alarm time t_a, refresh Δ, deadline D, threshold γ

$t_{\mathrm{ref}} \leftarrow t_a - \Delta_{\mathrm{ref}}$; $R \leftarrow \boldsymbol{x}_{t_{\mathrm{ref}}-w:t_{\mathrm{ref}}}$; $A \leftarrow \mathbf{0} \in \mathbb{R}^L$

for $t = t_a, t_a + \Delta, \ldots$ **while** $t + \ell \le t_{\mathrm{on}} + D$ **do**

  Build the 2^L hybrid segments Z_t^S; evaluate $e(\cdot)$ in one batch

  $\boldsymbol{\phi}_t \leftarrow$ Eq. (4); $A \leftarrow A + \omega_t \boldsymbol{p}_t$ (Eq. (5))

  Publish ranking argsort$(-A)$ and margin c; **if** $c \ge \gamma$ **then** publish H-LX tags for the top loop

end for

2. *Persistence versus oscillation.* A ridge model fitted on normal training data predicts each tag from the other tags of its own and adjacent loops. From its residuals over the last four samples, PI-CAX computes a persistence m_j (absolute mean, normalised) and an oscillation ratio $o_j = m_j / \overline{|e_j|}$, which is close to one for a one-sided shift. The decoupled evidence $z^c = 0.45\, z^{\mathrm{det}} + 0.55\, m\,(0.5 + 0.85\, o)$ is raised for actuators with a persistent shift and lowered for oscillating sensors of the same loop.
3. *Precision weighting and loop gating.* Relevance $\varphi = 0.6\,(z^c)^2 \operatorname{diag} \Lambda + 0.4\,|z^c \odot \Lambda z^c|$ uses the precision matrix Λ of the normal one-step residuals. It is scaled by the sign coherence of the residuals, and every loop is down-weighted by its energy relative to the most active loop.
4. *Operating envelope.* Tags outside their normal training range, or drifting monotonically within the window, are up-weighted.
5. *Counterfactual check.* For the five top-ranked tags, the last three samples are replaced by their loop-model predictions. Tags whose replacement lowers the detector score are reinforced.

Tag relevances are sparsified, normalised and summed per loop, and the loop map takes the role of PI-CAX's process stages. PI-CAX needs 16 detector evaluations and no gradients. Unlike LX, it satisfies no attribution axiom and depends on more than 30 hand-set constants. We include it to test whether physical priors help where score attribution alone fails.

5. Experimental design

Research questions. RQ1: Which explainers localize the attacked loop at the alarm, and does control-loop structure matter beyond grouping per se? RQ2:

How does accuracy evolve after the alarm, and does TEA help? RQ3: Which explainers deliver correct localizations before a deadline once detection delay and latency are accounted for? RQ4: How robust are they to sensor noise and a decoy manipulation? RQ5: How much does a random window split inflate detection results? Hypotheses H1–H6 about LX and TEA and the decision rule (two-sided tests, Holm-corrected $p < 0.05$ with the predicted sign) were fixed in the project's design document before the main runs (Supplementary Section S1). The comparison with PI-CAX was added afterwards and is reported as a post hoc comparison; PI-CAX's design and constants were frozen before it was first run on either benchmark.

5.1. Benchmarks

HAI 21.03. HAI 21.03 (Shin et al., 2021) is a hardware-in-the-loop testbed coupling a boiler (P1), a turbine (P2) and a water-treatment process (P3) with a simulated power plant (P4). It records 79 tags at 1 Hz: three normal training files (9.5 days) and five test files with 50 attacks (25 single, 25 combined). Its documentation lists the target controller(s) and manipulated points of every attack.

- *Ground truth.* We transcribed this table and matched every entry to the label runs in the data (all start times agree within 2 min); three point names were mapped to their 21.03 columns. An independent transcription of the same table, produced separately within the project, agrees on the target loops and points of 49 of 50 attacks. The one disagreement (A401) is a row continuing across a page break; we keep its second scenario, as for every A4xx attack, and excluding A401 changes no reported HR@1 by more than 0.005.

- *Loop map.* Tags were assigned to loop groups from the tag descriptions before any test data was examined (Supplementary Section S3). Twenty-two tags are constant in normal training and removed, leaving $d = 57$ tags in $L = 9$ groups.

- *Hidden setpoints.* Manipulated setpoints are largely invisible in the recorded data: of 36 (attack, setpoint-tag) targets, 30 are exactly constant during the attack and only 2 vary beyond their range in the preceding 10 min. Tag-level localization of those points is impossible from the data; loop-level localization is not.

PCL-Sim. The simulator models a supply tank, a transfer valve, a heated tank, a centrifugal pump and a discharge valve under five PI loops (tank-1 level, transfer flow, tank-2 level via pump speed, tank-2 temperature, discharge pressure), with actuator dynamics, hydraulic and thermal coupling, sensor noise, random-walk disturbances and an operator schedule that changes setpoints every 15–45 min. It exposes $d = 22$ tags in $L = 6$ groups (five loops and a disturbance group). Seven attack primitives mirror HAI (SP, CV offset, PV bias, PV freeze, CV+PV-freeze, SP+PV-freeze, short CV pulses), applied to each loop at two magnitudes and four test seeds, for 280 one-hour episodes. Controllers and historian see the manipulated values while the true state is integrated separately; the hazard deadline D^{haz} is the time to the first violation of a safety limit (overflow or dry-run, temperature $> 85\,^{\circ}\text{C}$, pressure > 8.5 bar). 113 attacks are hazardous (median $D^{\text{haz}} = 301$ s, range 6–1039 s). Twelve 6-hour normal runs with disjoint seeds are used for training and validation.

5.2. Detectors and leakage control

Four forecaster families are trained on normal data only, with three seeds each and window $w = 32$: a linear autoregressive model (closed-form ridge), an MLP, a dilated 1D-CNN (cf. Kravchik and Shabtai, 2018) and a GRU. Normal files are split chronologically (80 % train, 20 % validation); standardisation comes from training data, and σ_j and θ ($q = 0.999$, $\alpha = 0.1$) from validation data only. No test labels, files or seeds are used for any fitting or hyperparameter choice.

5.3. Explainers

All explainers attribute the same score $e(Z_t)$; tag-level attributions are summed per loop.

- *Tag-level baselines:* raw squared residuals (RE; the "MSE" ranking of Fung et al., 2024), saliency (SM; Simonyan et al., 2014), integrated gradients from the reference (IG, 32 steps; Sundararajan et al., 2017), KernelSHAP (Lundberg and Lee, 2017) and LIME (Ribeiro et al., 2016) with 2^L ("eq") or 1024 samples, and the ensemble of Fung et al. (2024) (RE + SM + LIME in place of LEMNA, $\beta = 2.5$ for actuators).

- *Loop-structured:* LX, LX-bg (LX with an interventional background of 8 training windows) and PI-CAX (Section 4.5).

- *Structure controls:* exact Shapley over L correlation clusters (CLX) or a random partition with the loop group sizes (RPX).
- *ShaTS-sensor* (Franco de la Peña et al., 2026), with $m = 20\,|G|$ coalitions per group; its cost ($\approx 2 \cdot 20\,d^2$ evaluations) restricts it to two detectors at the alarm.

Every method is run single-shot and with TEA.

5.4. Protocol, metrics and statistics

Following Fung et al. (2024), explanations are computed at a best-guess timing (oracle window starting at onset) and at practical timing: from the first alarm every $\Delta = 5\,\mathrm{s}$ for $180\,\mathrm{s}$.

- *Detection:* point-wise precision, recall and F1 without point adjustment (Kim et al., 2022), event recall (alarm within the attack or 60 s after it), median delay and false alarms per hour.
- *Localization:* HR@k (a target loop among the top k) and normalised rank, against a uniformly random ranking as chance reference.
- *Faithfulness:* deletion AUC, restoring loops in ranked order to the training mean (lower is better).
- *Stability:* top-k agreement between consecutive explanations, and Jaccard agreement under Gaussian input noise ($\sigma = 0.1$).
- *Cost and timeliness:* median wall-clock latency on one CPU thread, detector evaluations per explanation, DCL@$k(D)$ and the in-time rate with $t_{\mathrm{act}} = 30\,\mathrm{s}$.

The unit of analysis is the attack: seeds are averaged per (attack, detector), and these pairs are the paired units. We report means with 95 % bootstrap confidence intervals and test with a Friedman omnibus and two-sided Wilcoxon signed-rank tests against a reference configuration, Holm-corrected (Demšar, 2006; Holm, 1979), with matched-pairs rank-biserial effect sizes r_{rb}. Experiments ran with Python 3.13 and PyTorch 2.15 on an Intel Xeon Processor at 2.10GHz; latencies use one thread (`torch.set_num_threads(1)`).

Table 2: Detection performance (mean±sd over 3 seeds; no point adjustment). Event recall: alarm within the attack or 60 s after it. FA/h: false-alarm runs per hour of attack-free data.

Data	Detector	P	R	F1	Event rec.	Delay [s]	FA/h
PCL-Sim	Linear	0.79	0.17	0.28	0.96	6	1.38
	MLP	0.81±0.06	0.07±0.04	0.13±0.07	0.76±0.03	226±176	0.31±0.03
	CNN	0.83±0.03	0.19±0.02	0.31±0.02	0.89±0.00	46±38	0.96±0.06
	GRU	0.77±0.02	0.42±0.01	0.54±0.01	0.93±0.00	7±1	0.89±0.05
HAI 21.03	Linear	0.79±0.00	0.60±0.00	0.68±0.00	0.92	6	0.42±0.03
	MLP	0.75±0.01	0.50±0.02	0.60±0.02	0.90	15±1	0.36±0.04
	CNN	0.74±0.01	0.48±0.01	0.58±0.01	0.89±0.01	14±0	0.34±0.01
	GRU	0.78±0.00	0.55±0.00	0.65±0.00	0.90	14±1	0.36±0.02

6. Results

6.1. Detection

On HAI, the four detectors detect 89–92 % of the 50 attacks at 0.34–0.42 false alarms per hour, with median delays of 6–15 s (Table 2). On PCL-Sim, event recall is 76–96 %; point-wise recall is low because the PI controllers partly compensate the manipulations. In ten HAI attacks the score crosses θ 1–4 s before the labelled onset, which suggests that documented onsets lag slightly; we keep the documented labels. Localization is evaluated on detected attacks (181 (attack, detector) units on HAI, 1014 on PCL-Sim); deadline metrics count missed detections as failures.

6.2. Localization at the alarm (RQ1)

Tables 3 and 4 report loop localization for all configurations.

- *PI-CAX is the most accurate explainer.* It ranks the attacked loop first in 0.93 of detected attacks on HAI and 0.89 on PCL-Sim, significantly better than every other configuration (Holm $p < 10^{-5}$ on HAI and $p = 0.032$ on PCL-Sim, $n = 181$ and 1014).

- *Among score attributions,* LX is the most accurate on HAI (0.79; IG 0.75, Holm $p = 0.052$), whereas on PCL-Sim most score attributions lie within two points of each other (LX 0.85, LX-bg 0.87). H1 is therefore only partially supported.

- *Chance and structure.* A random ranking would rank a target loop first in 0.17 of the same units on both benchmarks (HAI 0.17, PCL-Sim 0.17). Exact Shapley values over correlation clusters (CLX) or

Table 3: Loop localization on HAI 21.03 (pooled over 4 detectors; mean over attacks, seeds averaged). Bold: best in column. Single-shot unless marked TEA.

	HR@1						HR@3	nRank
Method	best-guess	alarm	+60 s	+60 s TEA	+180 s	+180 s TEA	+60 s TEA	+60 s TEA
RE	0.49	0.71	0.48	0.61	0.35	0.57	0.89	0.09
SM	0.43	0.43	0.32	0.44	0.22	0.44	0.67	0.27
IG	0.50	0.75	0.56	0.72	0.33	0.70	0.93	0.06
KernelSHAP-eq	0.47	0.70	0.50	0.64	0.22	0.60	0.90	0.09
KernelSHAP-1024	0.48	0.74	0.51	0.69	0.28	0.65	0.92	0.07
LIME-eq	0.49	0.75	0.55	0.70	0.26	0.66	0.93	0.07
LIME-1024	0.49	0.76	0.56	0.69	0.31	0.66	0.93	0.07
Fung ens.	**0.60**	0.76	0.56	0.72	0.34	0.71	0.92	0.07
PI-CAX	–	**0.93**	**0.80**	**0.93**	**0.46**	**0.85**	**1.00**	**0.01**
CLX (clusters)	0.31	0.45	0.39	0.38	0.20	0.38	0.79	0.18
RPX (random)	0.17	0.25	0.19	0.21	0.16	0.17	0.68	0.26
LX-bg	0.55	0.74	0.58	0.71	0.31	0.65	0.94	0.06
LX	0.47	0.79	0.60	0.70	0.30	0.64	0.94	0.06

Table 4: Loop localization on PCL-Sim (pooled over 4 detectors; mean over attacks, seeds averaged). Bold: best in column. Single-shot unless marked TEA.

	HR@1						HR@3	nRank
Method	best-guess	alarm	+60 s	+60 s TEA	+180 s	+180 s TEA	+60 s TEA	+60 s TEA
RE	0.43	0.86	0.44	0.80	0.41	0.80	0.96	0.07
SM	0.44	0.65	0.43	0.70	0.40	0.69	0.95	0.10
IG	**0.68**	0.86	0.66	0.84	0.55	0.83	0.96	0.06
KernelSHAP-eq	0.46	0.80	0.47	0.71	0.40	0.69	0.90	0.13
KernelSHAP-1024	0.50	0.84	0.50	0.80	0.43	0.77	0.93	0.09
LIME-eq	0.42	0.82	0.46	0.72	0.40	0.70	0.91	0.12
LIME-1024	0.50	0.85	0.52	0.82	0.45	0.81	0.95	0.07
Fung ens.	0.48	0.87	0.49	0.82	0.45	0.82	0.97	0.06
PI-CAX	–	**0.89**	**0.79**	**0.87**	**0.72**	**0.87**	**0.99**	**0.04**
CLX (clusters)	0.12	0.27	0.16	0.23	0.14	0.23	0.78	0.31
RPX (random)	0.35	0.54	0.35	0.58	0.32	0.58	0.90	0.16
LX-bg	0.54	0.87	0.60	0.85	0.50	0.83	0.96	0.06
LX	0.57	0.85	0.53	0.82	0.46	0.80	0.95	0.07

random groups of the loop sizes (RPX) reach only 0.45/0.25 on HAI and 0.27/0.54 on PCL-Sim (all Holm $p < 10^{-10}$ against LX), so the gain of loop attribution comes from control structure, not grouping per se (H4 supported).

- *Best-guess timing* (Fung et al., 2024) does not help: on HAI it is worse than practical timing for every method except saliency, which is unchanged (LX 0.47), because many manipulations take tens of seconds to affect the process.

6.3. After the alarm: propagation and temporal accumulation (RQ2)

Single-shot score attributions degrade quickly after the alarm, for every method (Fig. 2). On HAI, LX falls from 0.79 at the alarm to 0.60 after 60 s and 0.30 after 180 s; on PCL-Sim from 0.85 to 0.53 and 0.46. Once controllers react, the largest residuals move to loops that compensate for the manipulated one, which adds a mechanism to the timing effect of Fung et al. (2024).

- TEA *recovers part of the loss for every method* (H2 supported). LX+TEA reaches 0.70 (HAI) and 0.82 (PCL-Sim) at 60 s (Holm $p = 0.011$ and $p = 7.9e - 56$ against single-shot), and single-shot top-1 outputs that change between consecutive explanations in 12–48 % of cases become 95–100 % consistent (Table 10). With TEA, the leading score attributions are within a few points of each other: on HAI LX is not significantly different from IG, LX-bg, the Fung ensemble, LIME-1024 or KernelSHAP-1024; on PCL-Sim IG and LX-bg are slightly but significantly better (Supplementary Table S2).

- *PI-CAX degrades far less.* Single-shot, it still reaches 0.80 (HAI) and 0.79 (PCL-Sim) at 60 s; with TEA, 0.93 and 0.87, significantly better than every other configuration (Holm $p < 10^{-6}$ and $p = 0.023$). With TEA, the attacked loop is among PI-CAX's top three in 1.00 and 0.99 of cases.

6.4. Why the physical prior works

Table 5 switches PI-CAX's components off one at a time (seed 0, all four detectors). Removing the persistence–oscillation decoupling reduces it to the level of the score attributions (HAI: 0.79 at the alarm and 0.59 single-shot at 60 s). Removing the precision weighting and loop gating, the operating envelope or the counterfactual check changes accuracy by at most 0.04. The advantage therefore comes from one physical hypothesis: the manipulated loop departs *persistently* from its normal cross-loop relations, while compensating loops *oscillate*. Score attributions cannot express this distinction, because they only ask how much each loop contributes to the current score.

Hard-to-explain attacks. Table 6 stratifies HAI by the type of manipulated variable, read from the target-controller code of the official attack table (strata of 3–23 attacks; descriptive only). Controller-output (CV) attacks move a recorded actuator and are localized well by most methods (0.79 at the alarm

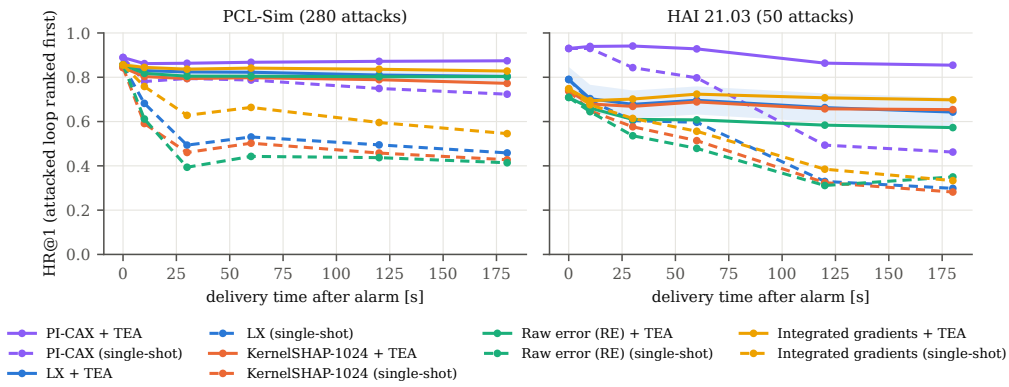


Figure 2: Top-1 loop localization versus delivery time after the alarm (solid: with TEA; dashed: single-shot; band: 95 % bootstrap CI for LX+TEA). Score attributions decay as the attack propagates through coupled loops; PI-CAX decays least.

Table 5: PI-CAX component ablation: loop HR@1 (single-shot at the alarm / single-shot at +60 s / +60 s with TEA; seed 0, four detectors pooled).

Configuration	HAI 21.03	PCL-Sim
Full PI-CAX	0.93 / 0.80 / 0.94	0.90 / 0.79 / 0.88
– persistence/oscillation decoupling	0.79 / 0.59 / 0.70	0.85 / 0.55 / 0.80
– precision weighting and loop gating	0.94 / 0.81 / 0.92	0.90 / 0.81 / 0.89
– operating envelope	0.90 / 0.80 / 0.91	0.90 / 0.78 / 0.89
– counterfactual check	0.93 / 0.79 / 0.93	0.90 / 0.79 / 0.88
none (smoothed residuals only)	0.75 / 0.52 / 0.60	0.86 / 0.47 / 0.81

for IG, KernelSHAP-1024, LX-bg and LX). Setpoint (SP) attacks are the hardest for score attributions: the setpoint is largely invisible in the data, and every score attribution falls to 0.38–0.52 at 60 s even with TEA. PI-CAX localizes them in 0.98 of cases, plausibly because a hidden setpoint change produces exactly the persistent PV/CV departure it looks for. Its weakest stratum is SP attacks combined with a PV spoof (0.52 at 60 s, $n = 4$), possibly because the spoofed PV hides the persistent departure.

6.5. Deadlines, latency and cost (RQ3)

Tables 7 and 8 and Fig. 3 account for detection delay, explainer latency and the correctness of the delivered explanation.

- *HAI.* DCL@1 at $D = 60$ s is 0.72 for PI-CAX+TEA against 0.53 for LX+TEA and at most 0.54 for any other configuration (Holm $p < 10^{-7}$,

Table 6: HAI 21.03 loop HR@1 by manipulated variable type, read from the target-controller code of the official attack table (attacks detected by at least one detector; single-shot at the alarm / +60 s with TEA; pooled over 4 detectors). Strata are small; descriptive only.

Manipulated variable	attacks	RE	IG	KernelSHAP-1024	LX-bg	LX	PI-CAX
SP only	8	0.69 / 0.38	0.66 / 0.52	0.66 / 0.46	0.65 / 0.47	0.73 / 0.46	0.93 / 0.98
SP + PV spoof	4	0.50 / 0.31	0.56 / 0.65	0.58 / 0.60	0.65 / 0.62	0.71 / 0.62	0.75 / 0.52
CV only	8	0.69 / 0.74	0.79 / 0.79	0.79 / 0.78	0.79 / 0.84	0.79 / 0.79	0.99 / 0.98
CV + PV spoof	3	0.94 / 0.89	0.97 / 0.92	0.92 / 0.78	0.97 / 0.94	1.00 / 0.89	1.00 / 1.00
combined (2 scenarios)	23	0.73 / 0.66	0.77 / 0.76	0.75 / 0.74	0.74 / 0.73	0.80 / 0.74	0.93 / 0.96

$n = 200$). It is bounded by event recall, so PI-CAX delivers a correct loop for most detected attacks within a minute of onset. No score attribution differs significantly from LX in DCL@1 at 60 s (H3 not supported for DCL).

- *PCL-Sim.* The correct loop is ranked first before the hazard deadline minus a 30 s action reserve in 0.48 (LX+TEA), 0.49 (PI-CAX+TEA) and 0.50 (IG+TEA) of hazardous cases. No explainer improves this rate materially, because the failures are dominated by detection: of the 704 failing (attack, detector, seed) cases of LX+TEA, 603 (86 %) were detected too late for any explanation to be delivered. Conditional on delivery, the top-ranked loop is correct in 0.87 (LX), 0.88 (PI-CAX) and 0.90 (IG) of cases.

- *Latency.* LX costs 2^L evaluations per explanation (512 on HAI, 64 on PCL-Sim), i.e. 0.3–2.5 ms on PCL-Sim and 2.2–16.9 ms on HAI on one thread (Table 9); PI-CAX needs 16 evaluations (median 1.4–1.7 ms during the runs, Tables 7 and 8). Both are deterministic and need no gradients; IG is cheaper than LX but requires gradient access.

- *Slower hardware.* We did not measure on embedded devices. Multiplying every measured latency by an assumed factor of 10 or 100 changes DCL@1 at $D = 30$ s by at most 0.04 for all configurations except LX-bg on HAI (4096 evaluations), which drops from 0.51 to 0.28. At these model sizes, latency is not the binding constraint; detection delay is.

- *ShaTS-sensor* needs $\approx 2 \cdot 20\, d^2$ evaluations (129960 on HAI), 4.8 s for the CNN. At the alarm on HAI it localizes 0.67 of attacks against 0.73 for LX on the same incidents (CNN, seed 0, $n = 45$; linear: 0.83 vs. 0.85), at roughly 250–280 times the cost.

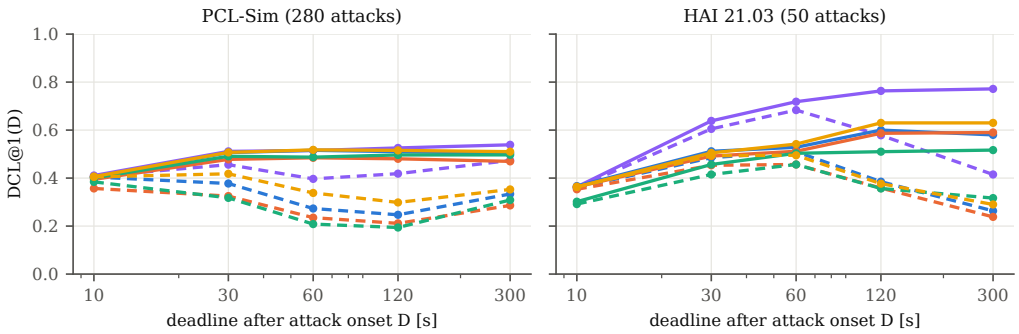


Figure 3: Deadline-conditioned localization DCL@1(D) (colours as in Fig. 2; solid: TEA; dashed: single-shot). Missed detections and undelivered explanations count as failures, so curves are bounded by event recall.

6.6. Explanation quality beyond accuracy

Table 10 reports further properties.

- *Faithfulness.* Deletion AUCs at the alarm are similar for all accurate methods (0.16–0.22), including PI-CAX; saliency and the structure controls are worse.
- *Stability under input noise.* On HAI, PI-CAX's single explanations are the most sensitive to Gaussian input perturbation (top-3 Jaccard 0.29 on HAI, against 0.56 for LX and 0.70 for KernelSHAP-1024), because its persistence statistics use only ten samples. Exact attributions are also less stable than fixed-seed sampling estimators. TEA mitigates both in operation.
- *Tags.* Within the top loop, H-LX ranks a manipulated tag first in 0.31 (HAI) and 0.45 (PCL-Sim) of attacks; PI-CAX's tag ranking reaches 0.55 and 0.57, the best of all methods.

6.7. Robustness (RQ4)

Table 11 summarises the robustness experiments (CNN detector, seed 0, unless noted).

Table 7: Deadline-conditioned localization on HAI 21.03 (T = with TEA). Latency: median measured per explanation during the runs (1 thread, pooled over detectors); Evals: detector evaluations per explanation.

Method	DCL@1(D)				DCL@3	Lat. [ms]	Evals
	30 s T	60 s	60 s T	120 s T	60 s T		
RE	0.46	0.46	0.50	0.51	0.68	0.5	1
SM	0.32	0.31	0.32	0.39	0.49	1.0	1
IG	0.51	0.49	0.54	0.63	0.68	3.4	32
KernelSHAP-eq	0.47	0.41	0.48	0.55	0.67	20.4	512
KernelSHAP-1024	0.49	0.46	0.51	0.59	0.68	40.9	1024
LIME-eq	0.50	0.48	0.52	0.60	0.68	38.1	512
LIME-1024	0.51	0.50	0.52	0.60	0.69	85.9	1024
Fung ens.	0.54	0.51	0.54	0.62	0.72	43.8	514
PI-CAX	**0.64**	**0.68**	**0.72**	**0.76**	**0.78**	1.7	16
CLX (clusters)	0.30	0.34	0.32	0.34	0.61	14.2	512
RPX (random)	0.16	0.17	0.19	0.17	0.52	13.0	512
LX-bg	0.51	0.51	0.53	0.60	0.70	200.2	4096
LX	0.51	0.51	0.53	0.60	0.70	28.2	512

- *Reference gap.* LX is insensitive to $\Delta_{\text{ref}} \in \{32, 60, 120, 300\}$ s on PCL-Sim. On HAI the default of 60 s is best at the alarm (0.73, against 0.64–0.71 for the other gaps), a difference of a few attacks; we did not tune the gap on test data. PI-CAX does not use the reference.

- *Sensor noise.* With the alarm threshold kept from clean data, Gaussian noise saturates the detector (precision 0.02 on HAI), so we recalibrate σ and θ on equally noisy validation data with frozen weights, as an operator would. Localization then degrades for the score attributions on HAI (LX+TEA 0.61 and 0.56 at $\sigma_n = 0.1/0.3$, against 0.67 without noise), and no score attribution differs significantly from LX (H5a not supported). PI-CAX, with its normalising statistics recalibrated on the same data, stays at 0.86 and 0.84. *Without* that recalibration it collapses (0.41 and 0.21), so its statistics must be refitted whenever sensor noise changes. On PCL-Sim, noise mainly delays detection, and all methods reach 0.91–0.95 at 60 s with TEA.

- *Decoy.* A small bias on the PV of an innocent loop captures the top rank in at most 0.05 of attacks for any method with TEA (PI-CAX 0.03). A non-adaptive decoy of this size is ineffective against all methods, so it cannot discriminate between them (H5b not supported).

Table 8: Deadline-conditioned localization on PCL-Sim (T = with TEA). Latency: median measured per explanation during the runs (1 thread, pooled over detectors); Evals: detector evaluations per explanation. In-time: hazard deadline minus a 30 s operator reserve, hazardous attacks only.

Method	DCL@1(D)				DCL@3	In-time@1		In-time@3	Lat. [ms]	Evals
	30 s T	60 s	60 s T	120 s T	60 s T	single	T	T		
RE	0.49	0.21	0.49	0.50	**0.53**	0.17	0.41	0.54	0.3	1
SM	0.47	0.21	0.46	0.47	**0.53**	0.19	0.40	0.54	0.8	1
IG	**0.51**	0.34	**0.52**	0.52	**0.53**	0.33	**0.50**	0.54	2.2	32
KernelSHAP-eq	0.40	0.23	0.40	0.40	0.48	0.25	0.40	0.51	2.4	64
KernelSHAP-1024	0.48	0.24	0.48	0.48	0.52	0.22	0.43	0.51	24.2	1024
LIME-eq	0.42	0.21	0.42	0.42	0.50	0.20	0.38	0.48	1.8	64
LIME-1024	0.50	0.24	0.50	0.50	**0.53**	0.22	0.46	0.52	15.4	1024
Fung ens.	0.50	0.23	0.50	0.50	**0.53**	0.21	0.44	0.54	3.2	66
PI-CAX	**0.51**	**0.40**	0.51	**0.53**	**0.53**	**0.40**	0.49	**0.55**	1.4	16
CLX (clusters)	0.10	0.09	0.10	0.10	0.41	0.10	0.14	0.38	1.6	64
RPX (random)	0.32	0.17	0.33	0.35	0.51	0.17	0.29	0.50	1.5	64
LX-bg	**0.51**	0.28	**0.52**	0.52	**0.53**	0.29	0.49	0.54	7.3	512
LX	**0.51**	0.27	**0.52**	0.51	**0.53**	0.28	0.48	0.53	1.6	64

6.8. Leakage from random window splits (RQ5)

A gradient-boosting classifier on window features (last value, mean and standard deviation of every tag) of the HAI attack files reaches F1 = 0.940 ± 0.008 with a random 70/30 split, but F1 = 0.575 ± 0.029 with a chronological, attack-disjoint split (train on test1–3, evaluate on test4–5; 5 seeds). Overlapping windows of the same attack instance on both sides of a random split inflate F1 by ≈0.37 (H6 supported); explanations of models evaluated this way inherit the optimism.

7. Discussion

What makes CPS explanations correct and timely?. Our results decompose the problem into four factors.

1. *Detection bounds timeliness.* On PCL-Sim, 86 % of the deadline failures are attacks detected too late or not at all, and no explainer can repair those. The deadline perspective puts detection latency (Puccetti and Ceccarelli, 2026) back at the centre.
2. *Closed-loop propagation defeats score attribution.* Every gradient-, sampling- or Shapley-based method degrades after the alarm, because controllers spread the effect of a manipulation to the loops that compensate for it. Explanations computed a minute after the alarm are therefore often misleading if read in isolation, consistent with the timing failure of Fung et al. (2024).

Table 9: Explanation latency on one CPU thread (Intel Xeon at 2.1 GHz, 4-core cloud VM; short unrelated jobs ran concurrently during part of the PCL-Sim run) (median over 10 incidents × 5 repetitions, ms; ShaTS: 2 incidents, 1 repetition). Evals: detector evaluations per explanation.

Data	Method	Evals	Linear	MLP	CNN	GRU
PCL-Sim	RE	1	0.1	0.1	0.3	0.6
	SM	1	0.4	0.6	0.9	3.7
	IG	32	0.5	1.0	3.1	5.7
	KernelSHAP-eq	64	1.2	1.5	3.1	2.8
	KernelSHAP-1024	1024	9.4	12.5	35.1	28.7
	LIME-eq	64	0.4	0.8	2.6	2.4
	LIME-1024	1024	2.0	5.1	26.6	21.0
	Fung ens.	514	1.9	3.7	15.4	15.1
	LX-bg	512	0.9	2.6	13.6	9.5
	ShaTS-sensor	19360	109.9	177.9	612.5	471.6
	LX	64	0.3	0.7	2.5	2.1
	H-LX (tags)	16	0.6	1.1	3.5	3.3
HAI 21.03	RE	1	0.1	0.2	0.3	0.7
	SM	1	0.4	0.7	0.9	3.7
	IG	32	0.7	1.4	3.5	6.1
	KernelSHAP-eq	512	6.8	10.2	20.9	18.7
	KernelSHAP-1024	1024	12.7	19.4	42.7	38.6
	LIME-eq	512	2.7	6.3	17.5	14.8
	LIME-1024	1024	4.6	12.0	33.5	31.2
	Fung ens.	514	3.3	7.6	18.6	19.7
	LX-bg	4096	14.1	44.3	165.0	185.2
	ShaTS-sensor	129960	1021.6	1871.4	4790.0	4384.8
	LX	512	2.2	5.9	16.9	13.7
	H-LX (tags)	64	2.8	6.6	20.0	17.2

3. *A physical prior removes most of this failure.* Separating persistent from oscillatory departures from normal cross-loop relations is a simple, plant-agnostic idea, and in our ablation it alone accounts for PI-CAX's advantage. On HAI it raised correct localization within 60 s of onset from 0.53 to 0.72, close to the ceiling of 0.78 set by detection (the share of attacks detected early enough for an explanation to be delivered); given delivery, PI-CAX ranked the attacked loop first in 0.92 of cases, against 0.68 for LX. It also handles hidden-setpoint attacks, which score attributions largely miss.
4. *Temporal accumulation is a cheap, general remedy.* TEA adds no detector evaluations, applies to any explainer, and turns flickering top-1 outputs into stable ones.

Table 10: Explanation quality. Temporal agreement: share of consecutive explanations (5 s apart, first 120 s) with the same top-1 loop. Noise Jaccard: top-3 overlap under Gaussian input noise ($\sigma = 0.1$) at alarm and +60 s. Deletion: area under the normalised score curve when loops are restored to the training mean in ranked order (lower is better), at the alarm. Tag HR@k: rank of a manipulated tag among all tags at the alarm (for LX: H-LX refinement); HAI excludes P2 setpoint targets that are invisible in the data.

Data	Method	Temporal top-1 agr.		Noise Jacc.@3	Deletion↓	Tag HR@1	Tag HR@3
		single	TEA				
PCL-Sim	RE	0.53	0.99	0.49	0.18	0.38	0.76
	SM	0.52	0.99	0.53	0.28	0.20	0.53
	IG	0.71	**1.00**	0.53	0.18	0.44	0.74
	KernelSHAP-eq	0.77	0.99	0.71	0.19	0.43	0.69
	KernelSHAP-1024	0.74	0.99	0.62	0.18	0.44	0.73
	LIME-eq	0.74	0.99	0.71	0.21	0.41	0.69
	LIME-1024	0.70	0.99	0.59	0.20	0.44	0.73
	Fung ens.	0.59	0.99	0.60	**0.17**	0.28	0.75
	PI-CAX	**0.88**	0.99	0.69	**0.17**	**0.57**	**0.92**
	CLX (clusters)	0.61	0.99	0.64	0.34	0.05	0.27
	RPX (random)	0.63	0.98	**0.76**	0.26	0.00	0.08
	LX-bg	0.66	0.99	0.55	0.22	–	–
	LX(H-LX)	0.64	0.99	0.51	0.21	0.45	0.76
HAI 21.03	RE	0.58	0.95	0.59	0.18	0.23	0.40
	SM	0.62	0.95	0.44	0.45	0.10	0.34
	IG	0.67	**0.97**	0.59	0.17	0.28	0.41
	KernelSHAP-eq	0.69	**0.97**	0.73	0.19	0.29	0.39
	KernelSHAP-1024	0.68	**0.97**	0.70	0.18	0.31	0.41
	LIME-eq	0.67	**0.97**	0.65	0.17	0.30	0.41
	LIME-1024	0.68	**0.97**	0.62	0.17	0.31	0.41
	Fung ens.	0.66	0.96	0.54	**0.16**	0.36	0.48
	PI-CAX	**0.82**	**0.97**	0.29	0.18	**0.55**	**0.70**
	CLX (clusters)	0.69	0.96	0.75	0.26	0.00	0.00
	RPX (random)	0.63	0.95	**0.76**	0.33	0.00	0.00
	LX-bg	0.68	0.96	0.57	**0.16**	–	–
	LX(H-LX)	0.68	0.96	0.56	0.17	0.31	0.49

Axiomatic versus physics-informed explanation. LX and PI-CAX represent two design philosophies. LX satisfies the Shapley axioms, has a cost fixed at design time and is deterministic; its weakness is that it faithfully attributes the *current* score, which after the alarm is dominated by compensating loops. PI-CAX satisfies no axiom and depends on more than 30 hand-set constants, but its one physical hypothesis matches how closed-loop processes respond to manipulation. That a heuristic designed on synthetic plants transferred to a real testbed without retuning is encouraging, but it also needs caution: it requires its normal statistics to be refitted when sensor noise changes, and its single explanations are sensitive to input perturbation. On the simulator its gains were small once evidence was accumulated, and a single real testbed cannot establish generality. Where gradients are available,

Table 11: Ablations and robustness (HR@1 unless noted; "alarm": single shot at the alarm; "+60 T": 60 s after the alarm with TEA). Reference-gap and noise rows: CNN detector, seed 0. Noise rows: Gaussian noise of standard deviation σ_n (standardised units) added to the test and validation data, with σ and θ recalibrated on the noisy validation data (model weights frozen; PI-CAX's normalising statistics are recalibrated on the same data). Decoy rows: linear and CNN detectors, 3 seeds; decoy@1 is the share of attacks in which the innocent decoy loop is ranked first.

Data	Condition	PI-CAX		LX		IG		KS-1024		RE	
		alarm	+60 T	alarm	+60 T	alarm	+60 T	alarm	+60 T	alarm	+60 T
HAI	Δ_{ref}=32 s	0.96	0.96	0.67	0.67	0.64	0.67	0.64	0.71	0.73	0.69
HAI	Δ_{ref}=60 s (default)	0.96	0.96	0.73	0.67	0.64	0.67	0.69	0.69	0.73	0.69
HAI	Δ_{ref}=120 s	0.96	0.96	0.71	0.64	0.69	0.67	0.69	0.67	0.73	0.69
HAI	Δ_{ref}=300 s	0.96	0.96	0.64	0.62	0.64	0.71	0.64	0.71	0.73	0.69
HAI	σ_n=0.1, recal.	0.82	0.86	0.64	0.61	0.61	0.70	0.66	0.68	0.64	0.73
HAI	σ_n=0.3, recal.	0.67	0.84	0.51	0.56	0.56	0.60	0.56	0.63	0.56	0.44
PCL-Sim	Δ_{ref}=32 s	0.86	0.84	0.79	0.76	0.78	0.75	0.78	0.78	0.82	0.79
PCL-Sim	Δ_{ref}=60 s (default)	0.86	0.84	0.80	0.76	0.80	0.77	0.79	0.78	0.82	0.79
PCL-Sim	Δ_{ref}=120 s	0.86	0.84	0.80	0.76	0.82	0.78	0.82	0.79	0.82	0.79
PCL-Sim	Δ_{ref}=300 s	0.86	0.84	0.81	0.78	0.82	0.78	0.82	0.78	0.82	0.79
PCL-Sim	σ_n=0.1, recal.	0.92	0.91	0.93	0.91	0.93	0.92	0.93	0.93	0.94	0.91
PCL-Sim	σ_n=0.3, recal.	0.94	0.94	0.94	0.93	0.95	0.95	0.95	0.94	0.95	0.93
PCL-Sim	no decoy: HR@1	0.87	0.86	0.84	0.80	0.84	0.82	0.83	0.77	0.85	0.78
PCL-Sim	decoy: HR@1	0.89	0.87	0.86	0.81	0.85	0.83	0.84	0.78	0.85	0.79
PCL-Sim	decoy: decoy@1	0.03	0.03	0.03	0.05	0.03	0.04	0.03	0.05	0.04	0.05

integrated gradients aggregated per loop with TEA is an equally strong and cheaper score attribution, and we recommend it as a baseline.

Practical recommendations.

1. Evaluate explanations at operational timing, against documented targets, and over the whole incident.
2. Group attributions by control loop, using the P&ID or controller configuration; tag-level ground truth for hidden setpoints cannot be met by any explainer.
3. Include persistence information, or a physical prior of comparable kind, in explanations of closed-loop processes, and accumulate evidence across the incident.
4. Present a short ranked list: with TEA the attacked loop was among PI-CAX's top three in 1.00 (HAI) and 0.99 (PCL-Sim) of detected attacks at 60 s.
5. Report deadline-conditioned metrics alongside latency, and use chronological, attack-disjoint evaluation.

7.1. Limitations and threats to validity

- *Benchmarks and CPS domains.* We evaluate on one real hardware-in-the-loop testbed (HAI 21.03) and one simulator of our own design.

Table 12: Outcome of the pre-specified hypotheses and of the post hoc PI-CAX comparison.

H	Statement	Outcome
H1	LX localizes better than tag-level methods	partially: best score attribution at the alarm on HAI; not better than IG, nor than LX-bg on PCL-Sim
H2	TEA improves accuracy after the alarm and temporal stability	supported (both benchmarks, all methods)
H3	LX+TEA has deterministic cost and the best DCL / in-time rate	cost: supported; DCL: not supported
H4	control structure beats data-driven / random grouping	supported ($p < 10^{-10}$)
H5a	LX+TEA degrades less under sensor noise	not supported
H5b	a decoy captures top-1 less often with LX+TEA	not supported (decoy ineffective against all methods)
H6	random splits inflate detection	supported (F1 0.940 vs. 0.575)
post hoc	PI-CAX vs. all other configurations	better at the alarm and at +60 s on both benchmarks and in DCL@1(60 s) on HAI; tied in the PCL-Sim in-time rate; needs recalibration under changed noise

SWaT and WADI are available only on request and could not be included; replication on them, newer HAI releases and TEP-based suites (Fung et al., 2024) is the most important next step. Both benchmarks are industrial process CPS; power grids, vehicles and medical devices also have control loops and deadlines, but we make no empirical claim about them.

- *Simulator bias.* A simulator designed by the authors of a method risks favouring it. All explainers query the same score function, attack primitives mirror HAI, and simulator parameters were fixed on development seeds before any explainer was evaluated.

- *Post hoc comparison.* PI-CAX was added after the hypotheses for LX were fixed. Its design and constants were frozen beforehand (repository history), and its adaptations to our interface are documented in the code, but it was not part of the pre-specified analysis plan.

- *Deadlines and hardware.* Hazard deadlines are computed in simulation without operator or safety-system response and with safety limits we chose; HAI provides no process safety times, so we report DCL over a grid of deadlines. Latencies were measured on one thread of a server CPU; edge-class conclusions rest on an assumed slowdown.

- *Method assumptions.* Both loop-level explainers need a loop map, and some HAI assignments are engineering judgement. Exact LX is exponential in L and needs a hierarchy beyond $L \approx 14$. Its incident-local reference assumes a normal state Δ_{ref} before the alarm. We study forecasting detectors only.

- *Adversary.* The decoy experiment is non-adaptive. Explanation-manipulation attacks that preserve the detector's decision while steering SHAP, LIME or integrated-gradient explanations have been demonstrated on security classifiers, including intrusion detection (Slack et al., 2020; Mia and Pritom, 2025). Whether an adversary can steer a loop-level or physics-informed explanation while respecting the constraints of a closed control loop remains open.

- *Human factors and literature search.* We measure whether the attacked loop is ranked first or within the top k, not whether operators act faster. Bibliographic APIs were not reachable from our environment, so the novelty check relied on the curated corpus and web search; this does not prove that no prior work exists.

8. Conclusion

We asked whether explanations of ML-based intrusion detectors in cyber-physical systems reach operators correctly and in time, and argued that XAI for CPS should be built on the control structure and evaluated against the physical deadlines of the process. With deadline-conditioned metrics, cross-checked loop-level ground truth for HAI 21.03 and a closed-loop simulator with hazard deadlines, we found that score attributions of every kind degrade as attacks propagate through coupled loops; that temporal accumulation recovers part of this loss for any explainer; that grouping by control loops is essential; and that a single physical prior, separating the persistent departure of the manipulated loop from the oscillatory compensation of the others, makes a physics-informed explainer significantly more accurate than all score attributions, both at the alarm and a minute later. Most remaining deadline failures are detection failures.

Future work should replicate the study on SWaT, WADI, newer HAI releases and other CPS domains; test physics-informed explainers against

adaptive adversaries who steer the explanation while keeping the alarm; measure latency on embedded hardware; and evaluate deadline-aware explanations with operators.

CRediT authorship contribution statement

Md Mazharul Islam: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Sazzad Hossain:** [roles to be confirmed by the authors].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

HAI 21.03 is publicly available from its maintainers (Shin et al., 2021). All code, configuration files, the HAI 21.03 loop-level ground-truth table and its independent cross-check, the PCL-Sim simulator, raw per-attack outputs and the scripts that generate every number, table and figure in this article are available at `https://github.com/mazharulmd/XAI-CPS`.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic), ChatGPT/Codex (OpenAI) and Gemini (Google) in order to write and refactor research code, run and analyse experiments, review the literature, and draft and edit the manuscript text. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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